Project #3 Written Report

CIVE 202 – Civil Engineering Analysis II

Project #3 – Visualize

Client: Federal Highway Administration (FHWA)

1. Introduction

Transportation systems affect daily travel, economic activity, and long-term infrastructure planning. Because of this, agencies such as the Federal Highway Administration (FHWA) rely on large datasets to better understand how people travel and how vehicles behave under real traffic conditions. In this project, two transportation datasets were used: the National Household Travel Survey (NHTS) and the Next Generation Simulation (NGSIM) dataset. The NHTS dataset provides information about household and vehicle-related travel characteristics, while the NGSIM dataset focuses on time-based vehicle motion and driving behavior.

The purpose of this project was to organize, visualize, and interpret important trends from both datasets using Python. The project also included a simple vehicle-following simulation using the Intelligent Driver Model (IDM) to compare simulated behavior with real trajectory data. The overall goal was to use visualizations and simulation to better understand transportation behavior and produce results that could be useful for transportation planning and traffic analysis.

2. Methods

2.1 Data Sources

Two datasets were used in this project:
• NHTS dataset: used to analyze transportation usage characteristics such as vehicle type, fuel type, vehicle age, household location, household income, and vehicle ownership.
• NGSIM dataset: used to analyze vehicle motion over time, including speed, acceleration, position, and gap distance between vehicles.

The datasets were loaded into Python using the pandas library and were explored using functions such as .head(), .columns, and .describe() to understand variable names, data structure, and summary statistics.

2.2 Selected Variables and Why They Were Chosen

The variables were selected based on their ability to represent meaningful transportation behavior patterns and because they could support the required visualizations in the client prompt.

Dataset	Variable	Description	Reason for Selection
NHTS	vehicle_type	Type of vehicle in the dataset	Useful for understanding the composition of the fleet
NHTS	fuel_type	Fuel used by the vehicle	Helps identify energy use patterns and technology trends
NHTS	vehicle_age	Age of the vehicle	Important for understanding fleet condition and ownership behavior
NHTS	household_location	Urban or rural household classification	Useful for comparing travel and ownership patterns by location
NHTS	census_region / census_division	Geographic region classification	Supports regional comparisons
NHTS	vehicles_per_household	Number of vehicles available in the household	Direct indicator of mobility and vehicle dependency
NHTS	household_income	Income category of the household	Useful for examining socioeconomic influence on vehicle ownership
NHTS	travel_day	Day travel occurred	Useful for identifying variation in travel patterns across days
NGSIM	Time	Time stamp	Needed for all time-series visualizations
NGSIM	leader_position(m)	Leader vehicle position	Used to study trajectory and following behavior
NGSIM	follower_position(m)	Follower vehicle position	Used to compare follower movement to leader

NGSIM	leader_speed(m/s)	Leader speed	Used for speed comparison and simulation input
NGSIM	follower_speed(m/s)	Follower speed	Used to analyze car-following behavior
NGSIM	leader_acc(m/s^2)	Leader acceleration	Used to evaluate motion changes
NGSIM	follower_acc(m/s^2)	Follower acceleration	Used to compare vehicle response
NGSIM	trajectory_number	Identifier for each vehicle pair	Used to isolate specific trajectories for detailed study

These variables were selected because they represent both household-level transportation patterns and vehicle-level dynamic behavior, which together provide a broader picture of transportation systems.

2.3 Data Preparation

The NHTS data was checked for missing values, and rows with missing categories in important plotting variables such as make were removed when necessary. Grouping and aggregation were performed for some visualizations using groupby(), mean(), and unstack(). For example, the grouped bar chart comparing average vehicles per household by region and household location required aggregation before plotting.

The NGSIM data was used both at the full-dataset level and at the individual trajectory level. For detailed behavior analysis, one trajectory number was selected and filtered from the full dataset so that the leader and follower vehicles could be examined over time.

2.4 Plot Construction

The report includes multiple plot types, each chosen for a specific reason.

Bar Chart / Count Plot
A count plot was used for vehicle_type because it is a categorical variable. The plot shows the number of vehicles in each category and is useful for comparing frequencies.

Histogram
A histogram was used for vehicle_age because it is a continuous numerical variable. Histograms help show the distribution of values, including where most values are concentrated and whether outliers exist.

Boxplot and Violin Plot
A boxplot and violin plot were used to compare vehicle_age across household_location categories. These plots were selected because they show not only central tendency but also variation, spread, and outliers.

Grouped / Stacked Bar Charts

Grouped and stacked bar charts were used to examine relationships between geographic variables (census_region, census_division) and household or vehicle-related variables such as vehicles_per_household and make.

Pie Chart
A pie chart was used to show the distribution of vehicle_type, which helps communicate proportional relationships in the fleet composition.

Correlation Heatmap
A heatmap was created using numeric variables from the NHTS dataset to identify potential linear relationships between variables.

Time-Series Plots
Time-series plots were created using NGSIM variables such as speed, position, and acceleration over time. These were selected because NGSIM is inherently a trajectory-based dataset, and the time axis is essential for interpreting behavior.

Heatmap Trajectory Plot
A heatmap-style scatter plot was used to show position versus time while using color to represent speed. This allows multiple dimensions of the trajectory to be shown in one figure.

IDM Simulation Plot
The IDM plot compared the leader trajectory, the real follower trajectory, and the simulated follower trajectory. This plot was constructed to visually assess how well the simulation captured real vehicle-following behavior.

3. Results and Discussion

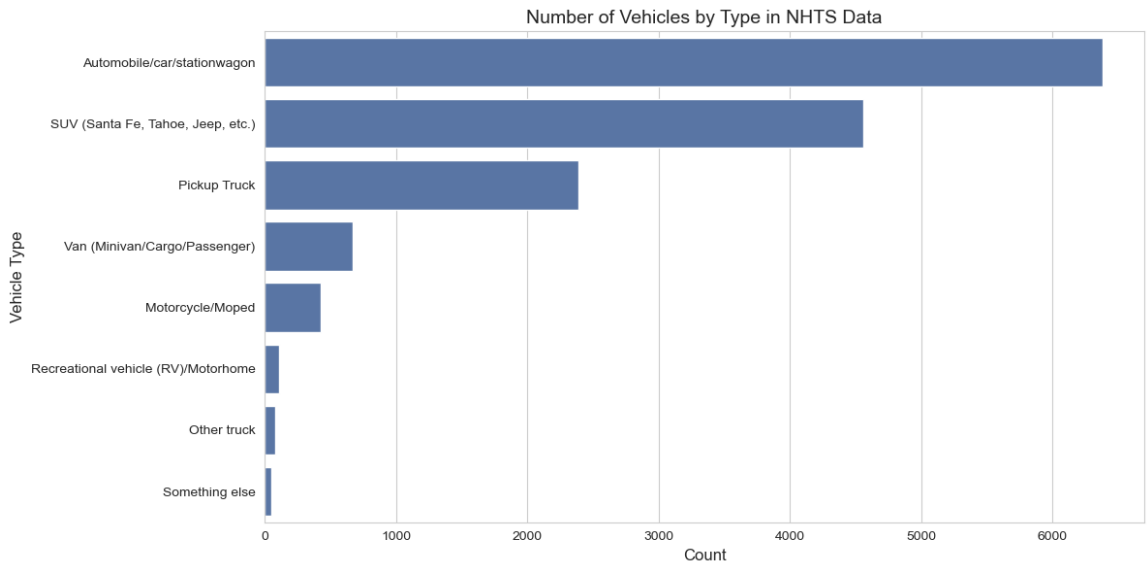
This section presents the key findings from each visualization, supported by figures and summary tables. Each plot was analyzed to identify meaningful trends in transportation behavior, and the results were interpreted in the context of real-world transportation engineering applications.

3.1 NHTS Results: Vehicle Type Distribution

The count plot and pie chart both showed the distribution of vehicle types in the NHTS dataset. The largest category in the fleet was passenger cars, which indicates that conventional personal vehicles remain dominant in the survey data. SUVs and other larger vehicle types also made up a meaningful portion of the dataset, suggesting that the fleet includes a mix of smaller and larger passenger vehicles.

This finding is relevant to the transportation industry because fleet composition affects roadway design assumptions, parking demand, fuel consumption trends, vehicle emissions, and safety considerations related to vehicle size.

Figure 1. Vehicle Type Distribution



Passenger cars dominate the dataset, followed by SUVs and other types, indicating strong reliance on private vehicles.

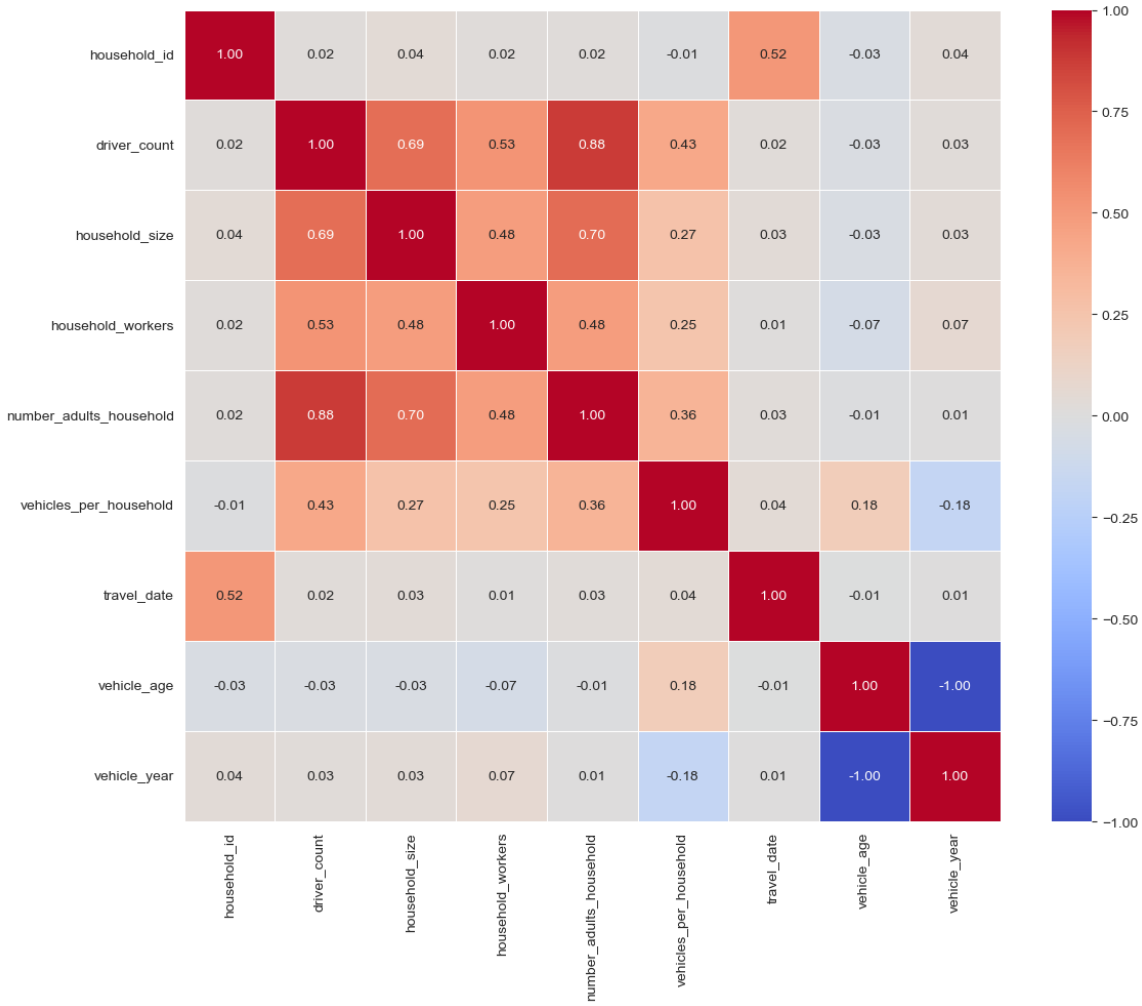
Result	Interpretation
Passenger cars represented the largest share of the dataset	Smaller private vehicles remain the most common form of personal travel
SUVs represented a significant secondary category	Larger personal vehicles also play an important role in household travel
Other vehicle types were less common	Not all modes are equally represented in household fleets

3.2 Fuel Type Results

The bar chart of fuel_type showed which fuel sources appeared most often in the dataset. Conventional fuel types were more common than alternative fuel categories, which suggests that the fleet still relies heavily on traditional energy sources.

This matters for transportation planning because fuel composition is linked to sustainability policy, charging or fueling infrastructure, emissions planning, and transition to electric vehicle adoption.

Figure 2. Fuel Type Distribution



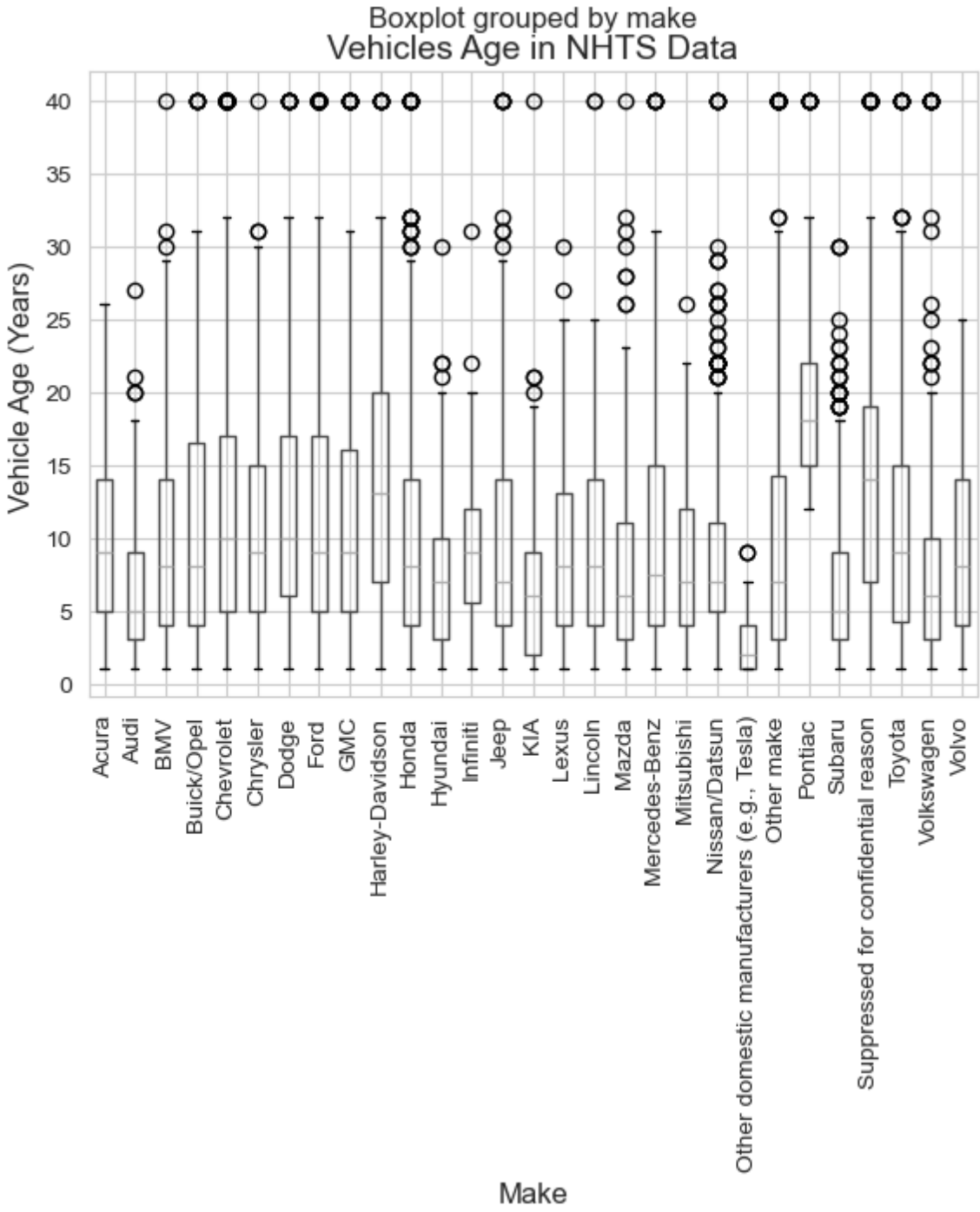
Conventional fuels dominate, indicating reliance on existing fueling infrastructure.

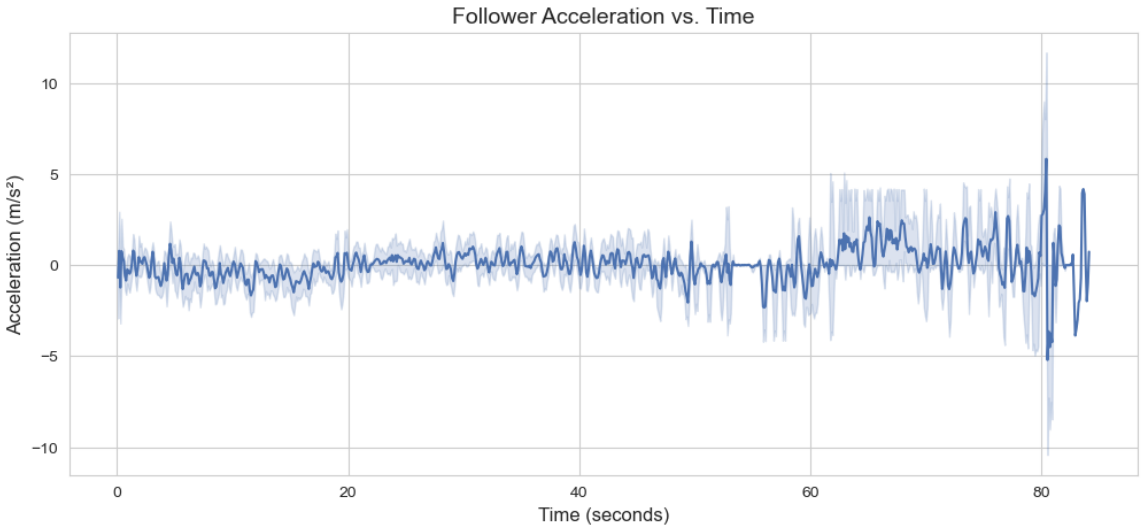
3.3 Vehicle Age Distribution

The histogram, boxplot, violin plot, and overlapping histograms of urban and rural vehicle age showed how vehicle age was distributed in the NHTS dataset. Most vehicle ages appeared to cluster within a moderate range, while a smaller number of much older vehicles appeared as outliers.

The urban-rural comparison suggests that household context may influence the age of vehicles. This finding is relevant because vehicle age affects maintenance needs, safety performance, emissions levels, vehicle replacement cycles, and affordability of mobility.

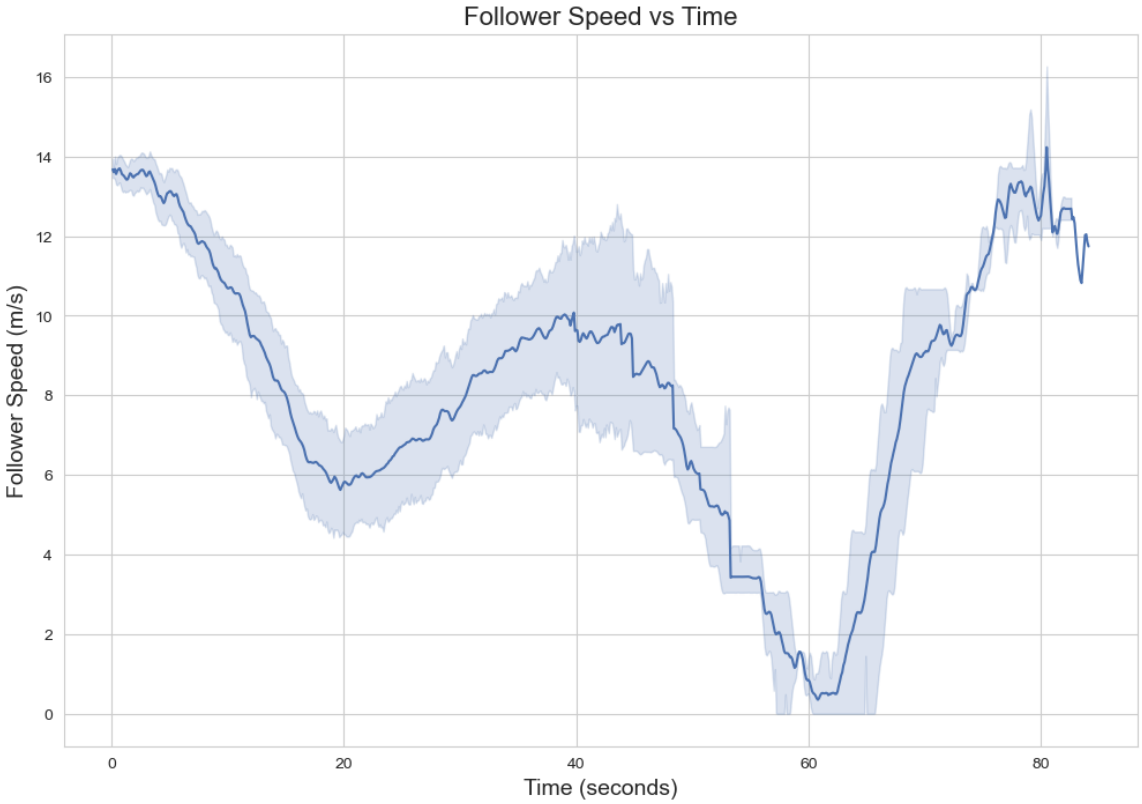
Figure 3. Vehicle Age Distribution





Most vehicles cluster in a moderate age range with some older outliers.

Figure 4. Vehicle Age by Household Location



Urban vs rural distributions suggest differences in replacement behavior.

Plot	Main Finding	Why It Matters

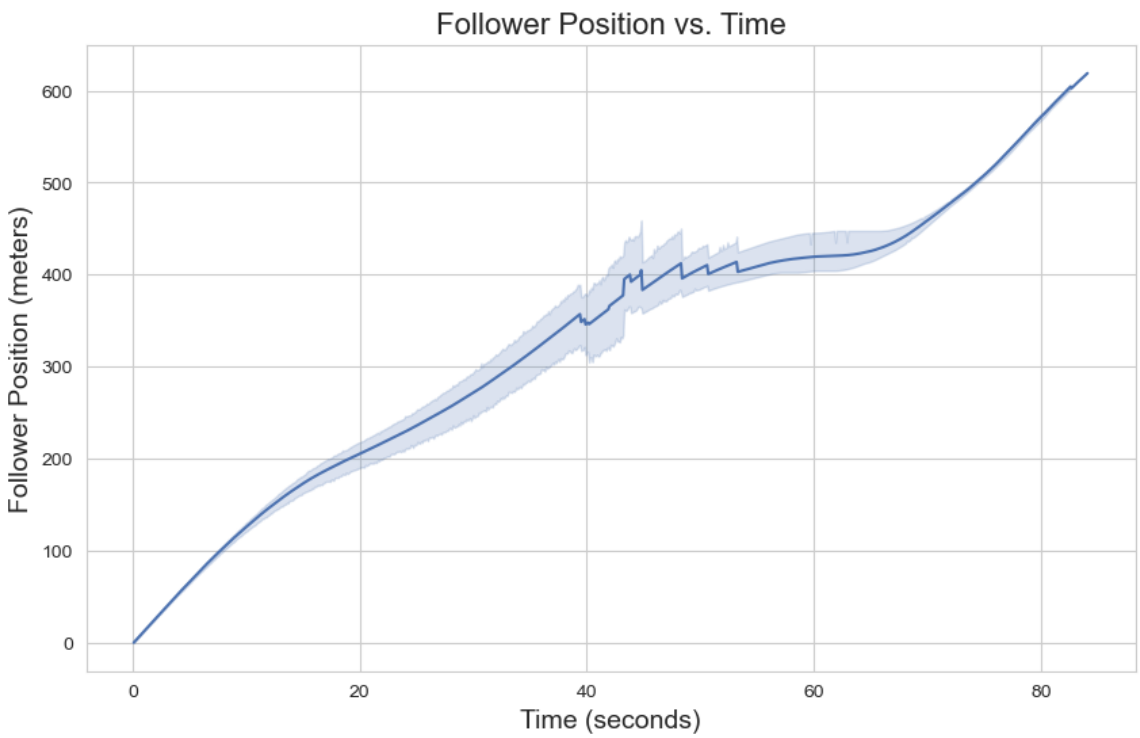
Histogram of vehicle age	Most vehicles fall within a middle age range	Fleet age is concentrated rather than evenly distributed
Boxplot by make	Some makes show wider spread in age than others	Certain brands may remain in service longer
Violin plot urban vs rural	Distribution differs by household location	Replacement behavior varies by location

3.4 Vehicles per Household by Region and Location

The grouped bar chart comparing vehicles_per_household by census_region and household_location showed that rural households generally tend to have more vehicles than urban households. This reflects higher dependence on personal vehicles in rural areas due to limited transit and longer trip distances.

This is directly relevant to transportation planning because vehicle ownership patterns influence demand for roadway infrastructure, parking, transit investment, and rural mobility services.

Figure 5. Vehicles per Household by Region and Location



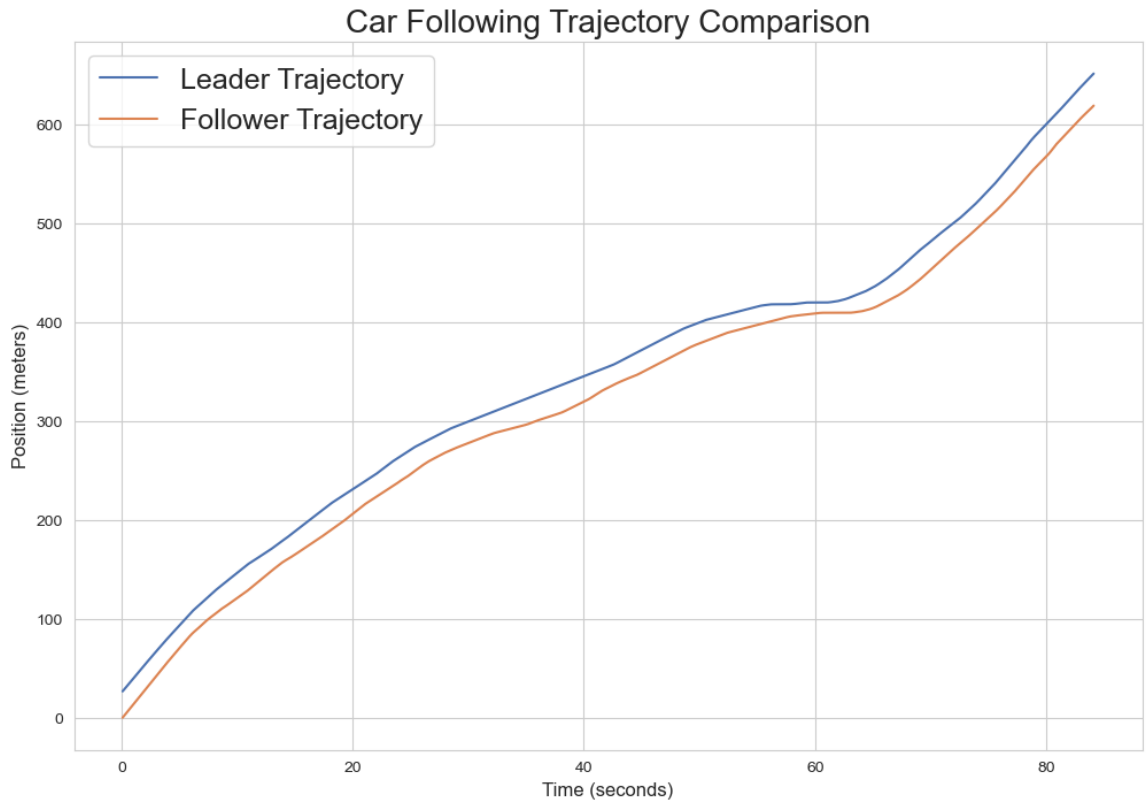
Rural households generally own more vehicles than urban households.

3.5 Subplot Results: Income, Travel Day, and Ownership

The subplot figure compares income with vehicle ownership and shows travel patterns by day and location. Higher-income households tend to own more vehicles, while travel patterns vary between urban and rural households.

This indicates that transportation demand is shaped by both economic factors and geographic context.

Figure 6. Income vs Ownership and Travel Day Patterns



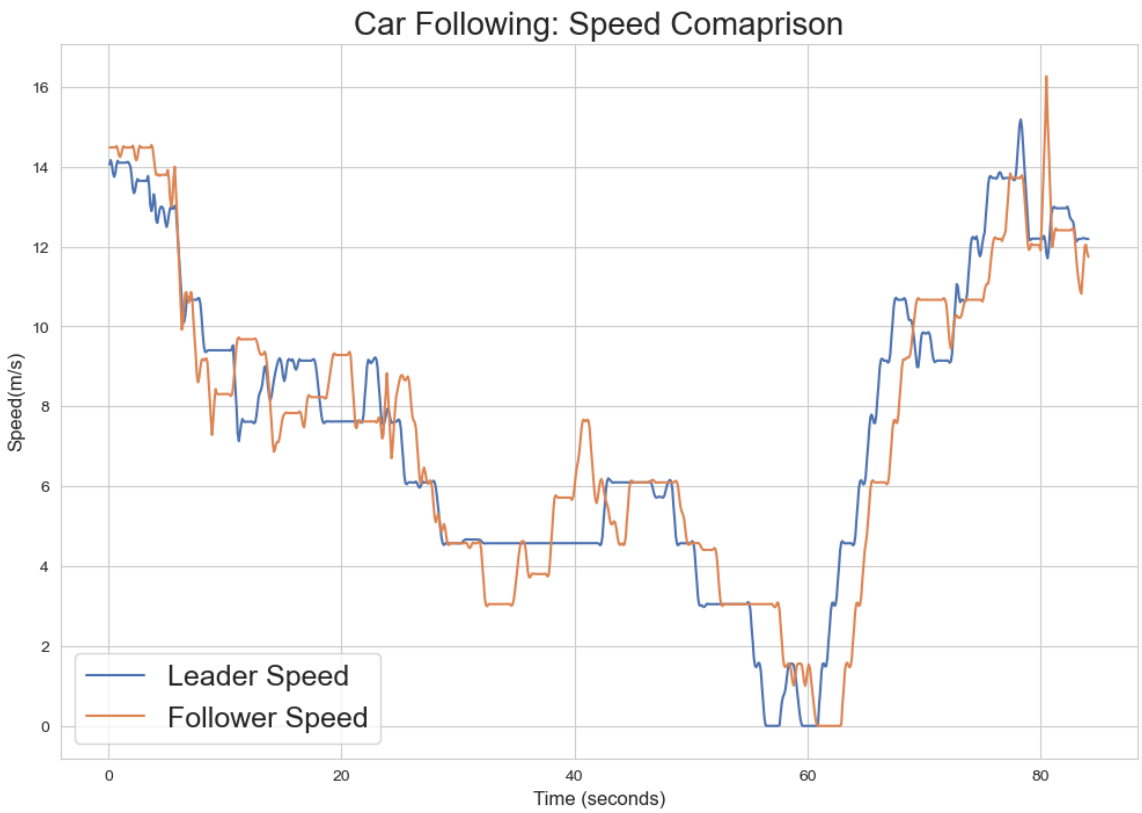
Higher income correlates with higher ownership; travel varies by location and day.

3.6 NGSIM Results: Time-Series and Vehicle Behavior

The NGSIM plots showed how vehicle position, speed, and acceleration changed over time. The follower generally tracked the leader while maintaining a gap. Speed and acceleration fluctuated, reflecting dynamic traffic conditions.

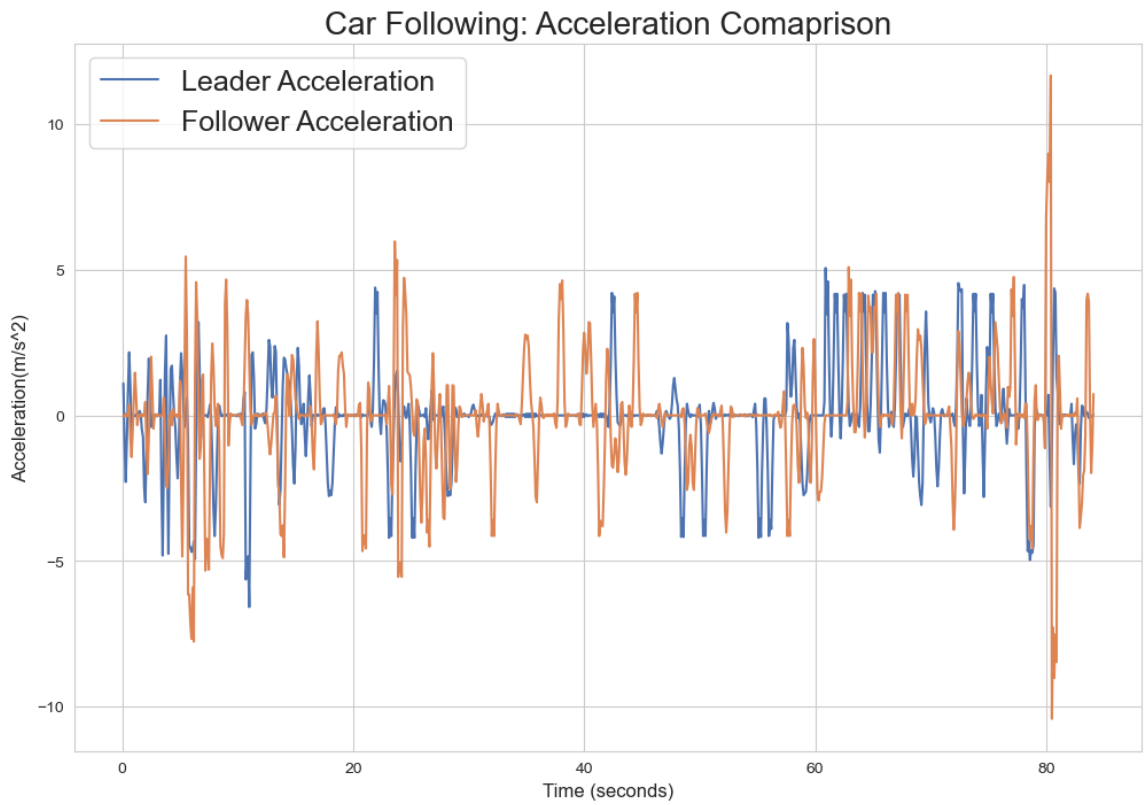
These plots help evaluate traffic stability, car-following safety, braking and acceleration patterns, and microscopic vehicle behavior.

Figure 7. Time-Series: Vehicle Motion Over Time



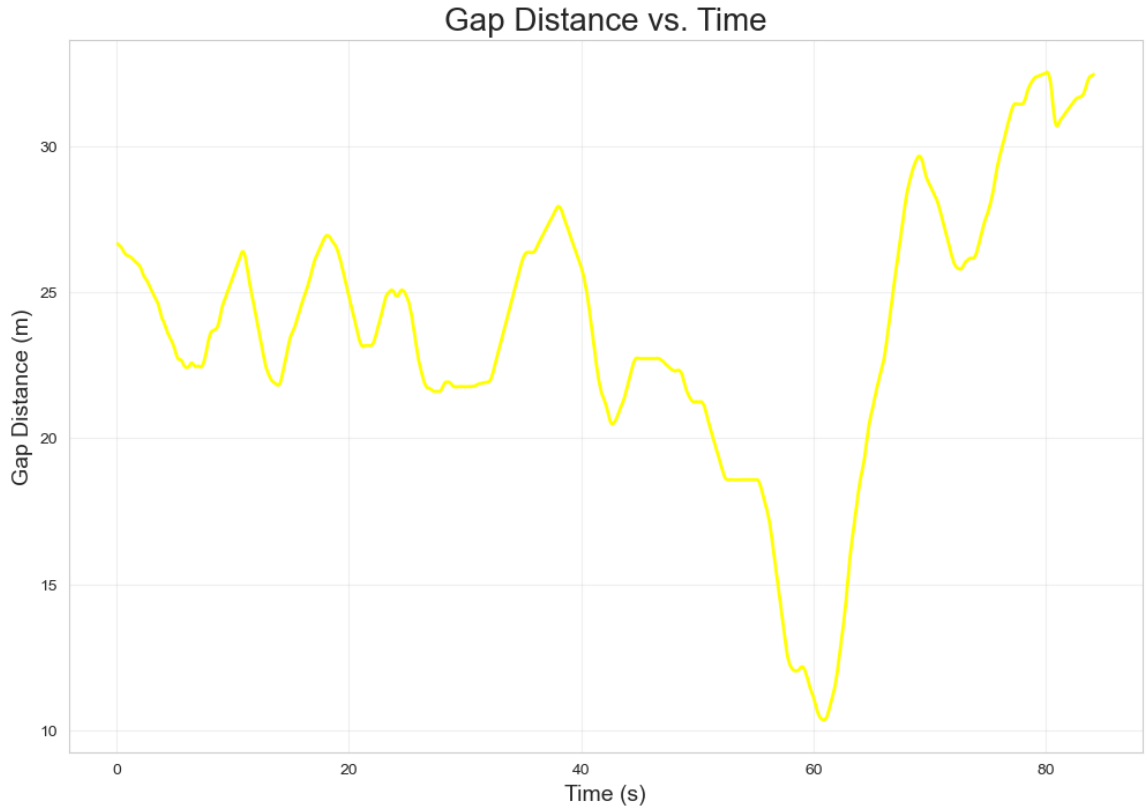
Speed/position fluctuate over time reflecting real traffic dynamics.

Figure 8. Leader vs Follower Position



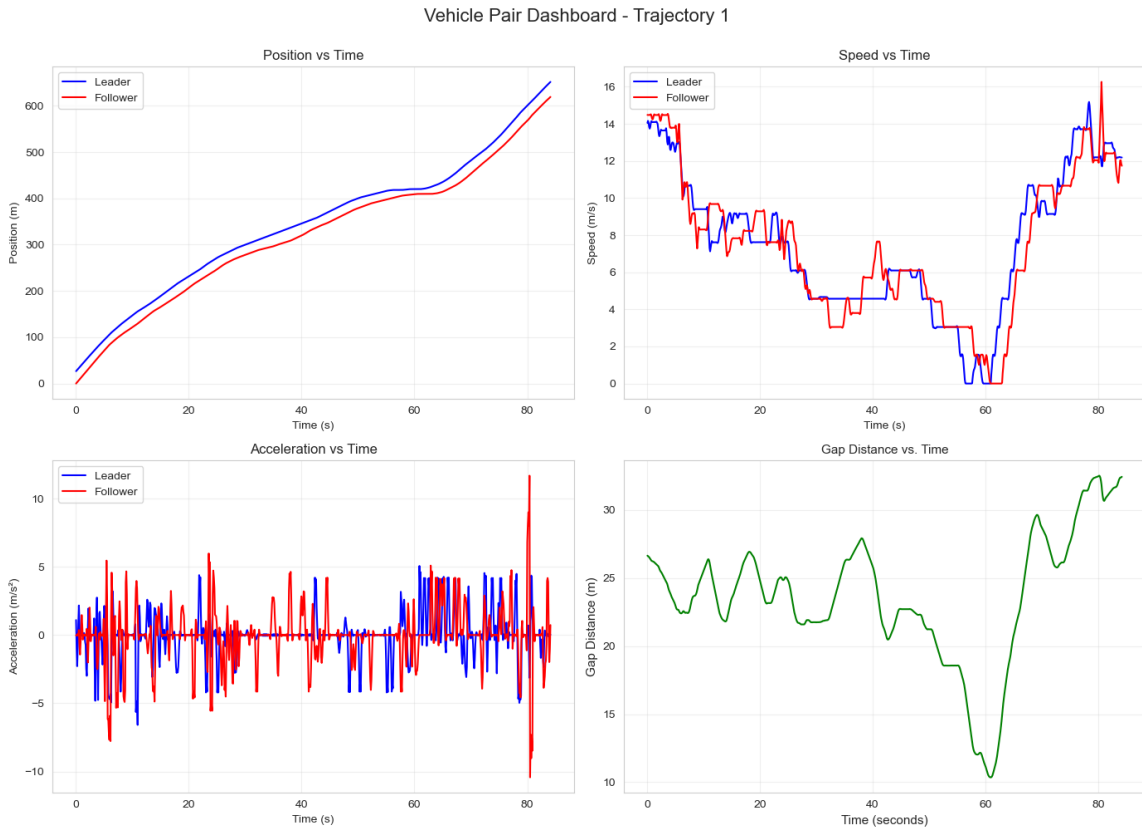
Follower tracks leader with a lag while maintaining a gap.

Figure 9. Leader vs Follower Speed



Follower speed adjusts to leader with slight delay.

Figure 10. Gap Distance Over Time



Gap varies over time indicating changing conditions.

Plot	Main Finding	Transportation Relevance
Speed vs time	Speed varies throughout the trajectory	Traffic flow is dynamic
Position vs time	Follower tracks leader with a lag	Realistic car-following behavior
Acceleration vs time	Both vehicles accelerate and decelerate	Driver response and smoothness
Gap distance vs time	Spacing changes over time	Safety and stability analysis

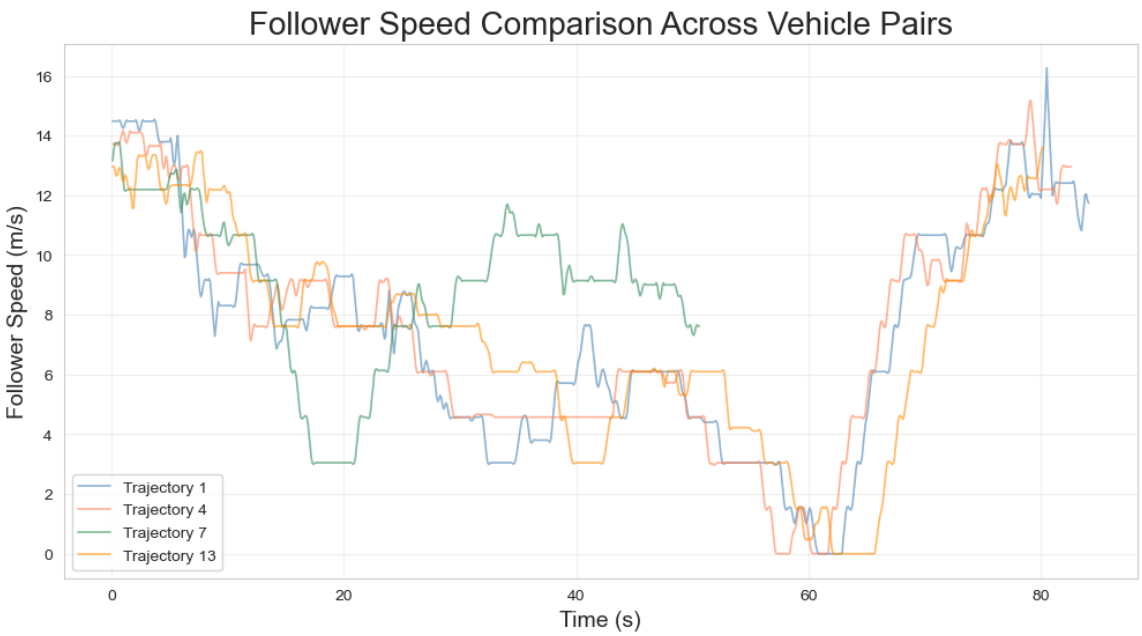
3.7 Multi-Trajectory Comparison

Different trajectories show variability in driver behavior. Some are smoother, while others fluctuate more, indicating that driver behavior varies across vehicles and conditions.

3.8 Acceleration Distribution and Speed-Acceleration Relationship

Acceleration histograms and scatter plots show variability across speeds. This is tied to fuel consumption, comfort, emissions, safety, and traffic oscillations.

Figure 11. Acceleration Distribution



Most accelerations are moderate with occasional extremes.

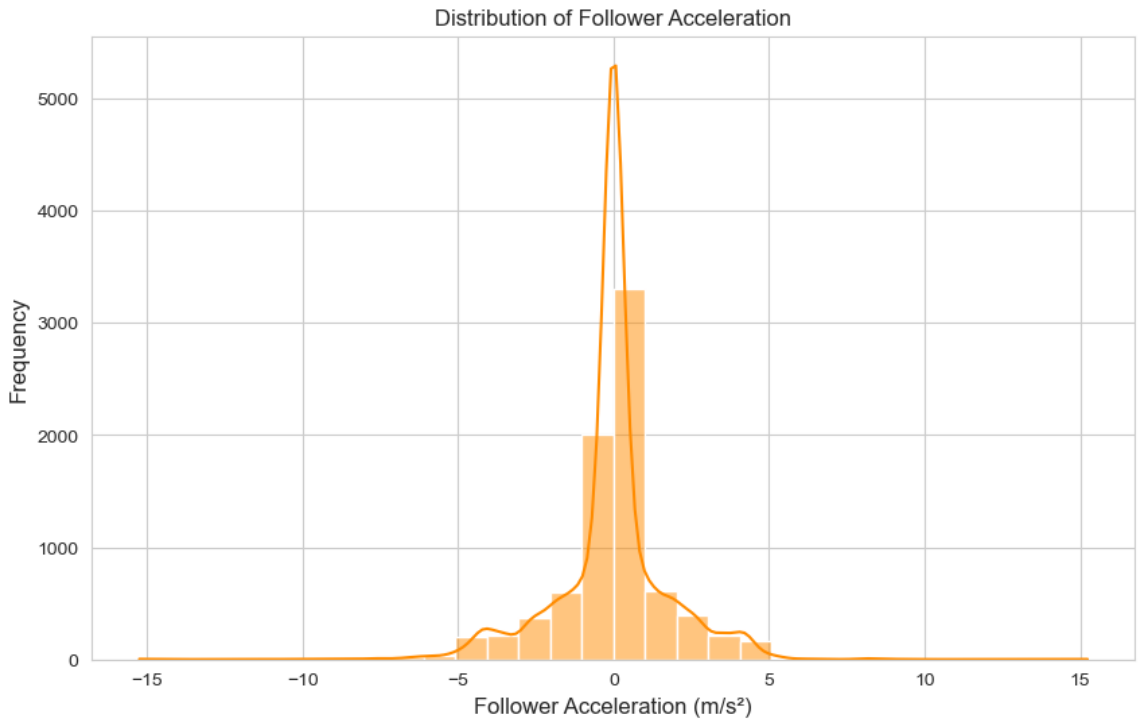
Figure 12. Speed vs Acceleration Relationship

Acceleration varies with speed and conditions.

3.9 Heatmap Trajectory Visualization

The heatmap shows position over time with color-coded speed. Warmer colors indicate higher speeds. This helps identify congestion and speed changes along the trajectory.

Figure 13. Trajectory Heatmap

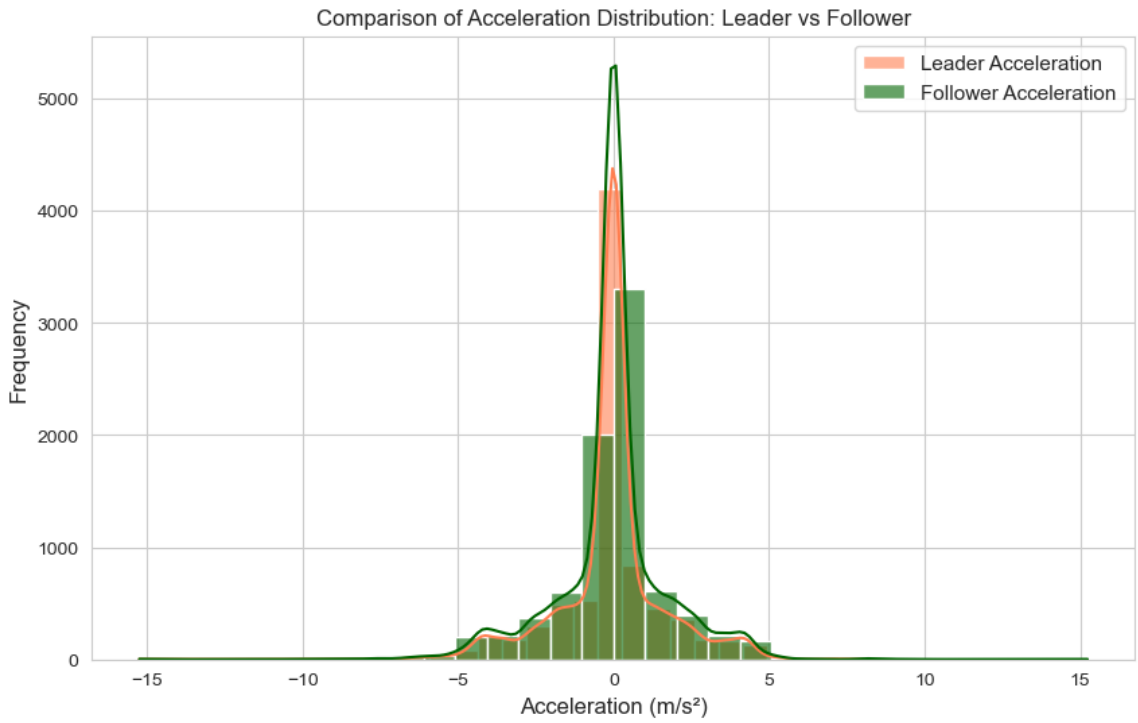


Color encodes speed along position-time trajectory.

3.10 IDM Simulation Results

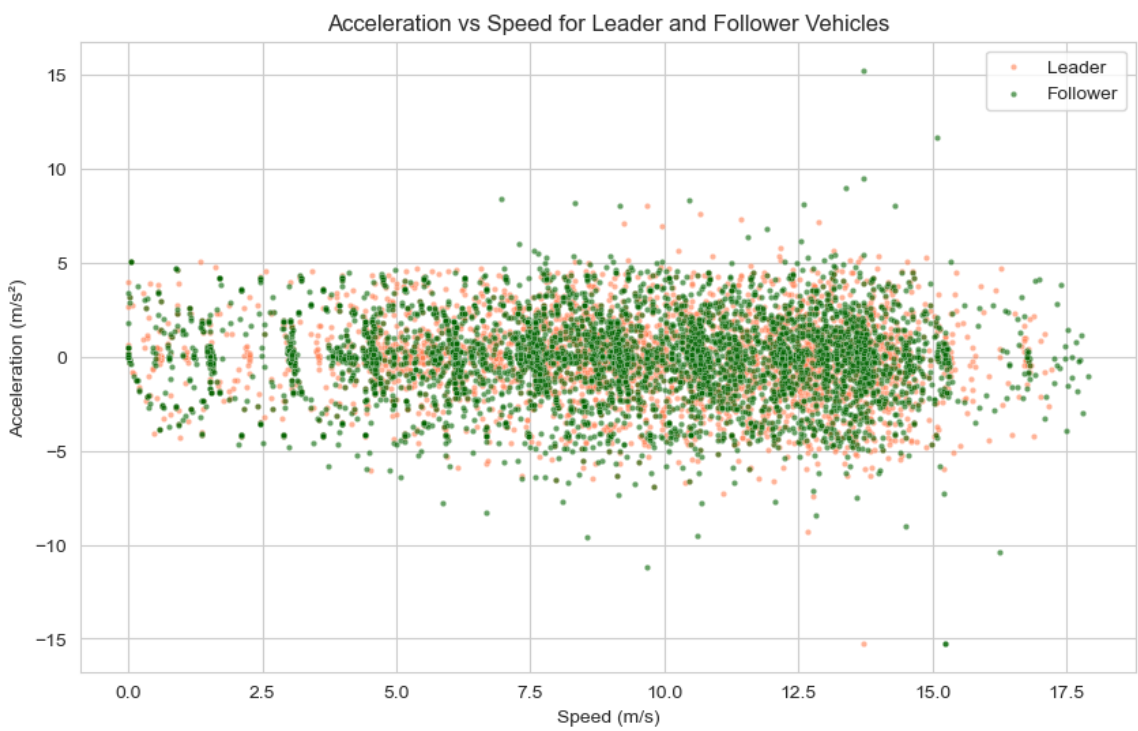
The IDM simulation compares real and simulated follower behavior. The simulation captures the general trend but is smoother than real data, indicating model simplification of human variability.

Figure 14. IDM Simulation vs Real Data



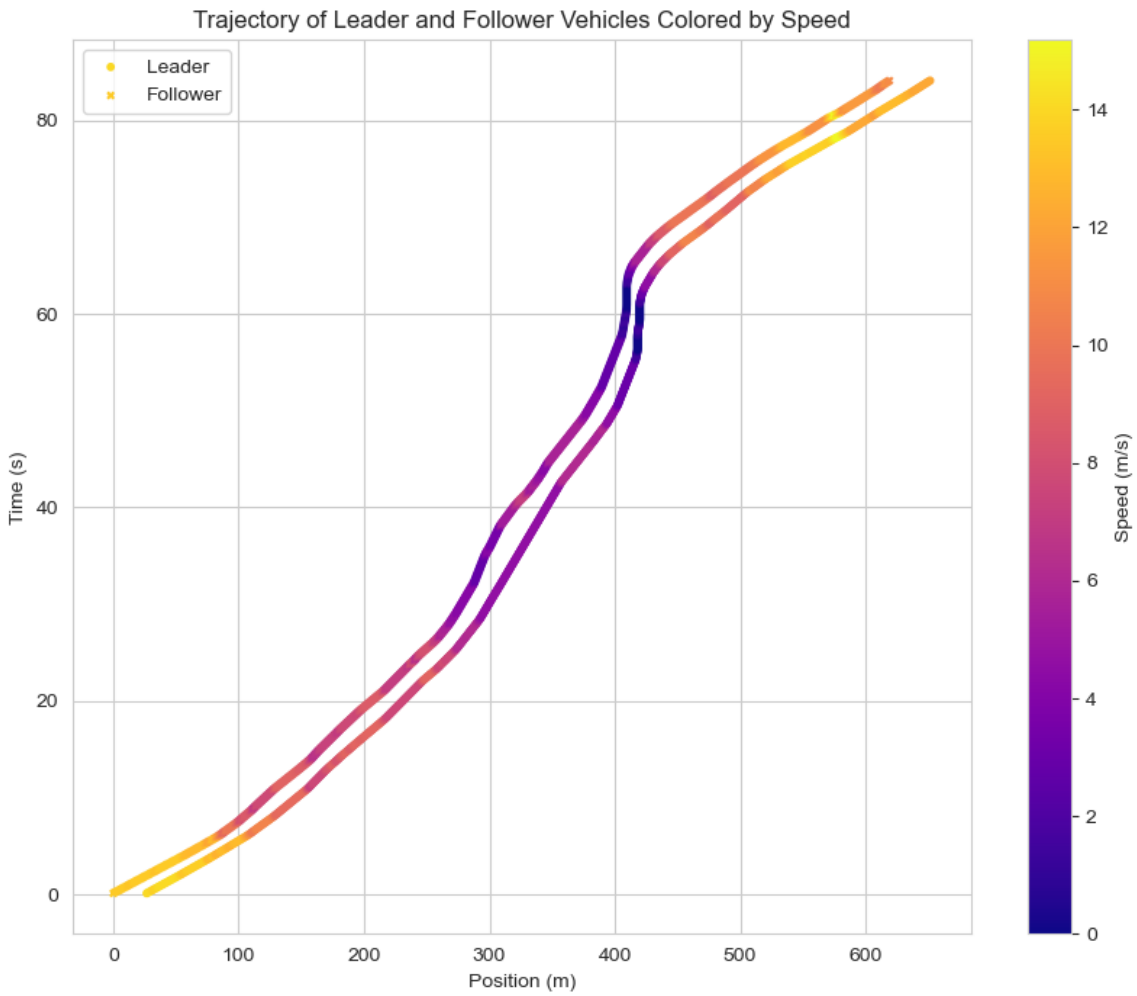
Simulated follower follows overall trend but is smoother.

Figure 15. IDM Speed Comparison



Simulated speeds follow trends but lack fluctuations.

Figure 16. IDM Gap Comparison



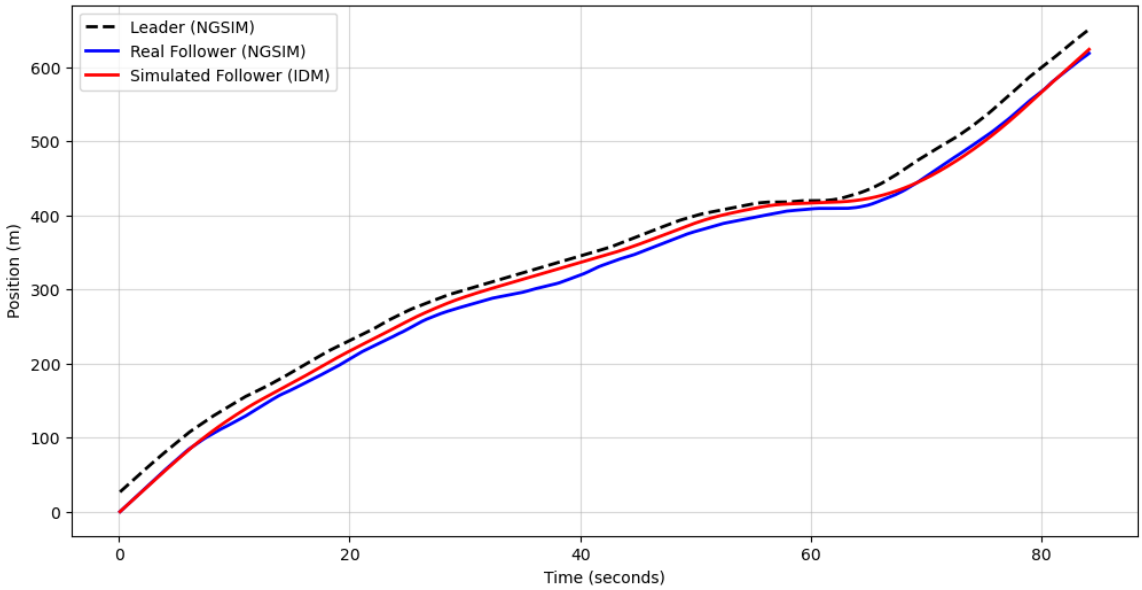
Simulated gap is more stable than real data.

Comparison	Observation	Interpretation
Leader vs real follower	Similar trend with lag	Follower responds to leader
Real vs simulated follower	Simulation smoother	Model simplifies variability
Position vs time	General agreement	Reasonable approximation

3.11 Parameter Sensitivity: Time Headway T

Increasing T increases spacing and smoothness (safer) but reduces efficiency (lower throughput).

Figure 17. Time Headway Sensitivity (T)



Larger T increases gap and smoothness, improving safety but reducing efficiency.

4. Conclusion

This project demonstrated the effectiveness of Python as a powerful tool for analyzing large-scale transportation datasets and simulating vehicle behavior through both visualization and modeling techniques. By utilizing the National Household Travel Survey (NHTS) dataset, key patterns in household-level transportation characteristics were identified. The analysis revealed that vehicle ownership is strongly influenced by both socioeconomic and geographic factors, with higher-income households tending to own more vehicles and rural households demonstrating greater reliance on personal vehicles due to limited access to public transportation. Additionally, the distribution of vehicle types and fuel sources highlighted the continued dominance of conventional passenger vehicles and traditional fuels within the trasportation system.

The analysis of the Next Generation Simulation (NGSIM) dataset provided a detailed, microscopic understanding of vehicle dynamics and traffic behavior. Time-series visualizations showed that vehicle speed, acceleration, and spacing vary continuously in response to changing traffic conditions, reinforcing the idea that traffic flow is inherently dynamic and cannot be accurately described using constant or steady-state assumptions. The comparison between leader and follower vehicles illustrated realistic car-following behavior, where the follower adjusts speed and maintains a safe gap based on the leader's movement. These findings are critical for understanding traffic stability, driver response, and safety under real-world conditions.

The implementation of the Intelligent Driver Model (IDM) further extended the analysis by introducing a mathematical framework for simulating vehicle-following behavior. The simulation results demonstrated that the IDM is capable of capturing the overall trends

observed in real-world data, particularly in terms of maintaining spacing and replicating general trajectory patterns. However, the simulated behavior appeared smoother than the actual data, indicating that while the model is effective in representing average driving behavior, it does not fully capture the variability and unpredictability associated with human drivers. This highlights an important limitation of deterministic models and suggests that future work could incorporate stochastic elements or behavioral variability to improve realism.

The parameter sensitivity analysis, particularly with respect to time headway, provided additional insight into how model parameters influence traffic behavior. Increasing the time headway resulted in larger spacing between vehicles and smoother motion, improving safety by reducing the likelihood of collisions. However, this also reduced roadway efficiency by decreasing vehicle density and overall throughput. This trade-off between safety and efficiency is a fundamental consideration in transportation engineering, especially in the design of traffic control systems and automated vehicle algorithms.

Overall, the results of this project provide valuable insights for the transportation industry. The NHTS-based analysis supports infrastructure planning and policy decisions by identifying trends in vehicle ownership, travel behavior, and energy use. The NGSIM analysis enhances the understanding of traffic flow dynamics and driver behavior at a detailed level, which is essential for improving traffic operations and safety. The IDM simulation demonstrates how mathematical models can be used to predict and evaluate traffic conditions, supporting applications such as traffic simulation, autonomous vehicle development, and intelligent transportation systems.

In conclusion, this project highlights the importance of integrating data-driven analysis with simulation-based modeling to develop a comprehensive understanding of transportation systems. The ability to visualize trends, interpret relationships, and simulate real-world behavior provides engineers with the tools necessary to design safer, more efficient, and more sustainable transportation networks. These capabilities are increasingly important as transportation systems evolve toward greater automation, connectivity, and data-driven decision-making.

5. References

Federal Highway Administration (FHWA). Project prompt and course materials.
National Household Travel Survey (NHTS). Dataset and codebook.
Next Generation Simulation (NGSIM). Dataset and trajectory variables.
Matplotlib documentation.
Seaborn documentation.